Nasr Kasrin

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/nkasrin

Beyond the maker/manager divide: I'm a maker as in engineering design, R&D innovation, and software architecture; a manager as in product and cross-functional leadership (*See n42r.github.io*)

Experience

2015 – 2021 **Research Associate**, *University of Bamberg (EU Project)*, Germany
- Played a key role in developing a data management solution for a project of 8 companies.
- Interfaced and extracted requirements from domain experts in manufacturing.
- Research and developed the data architecture and published several papers.
- Served as the single source of truth on the project.
- Led a team to develop software components.

2013 – 2015 **Assistant Dean / Lecturer**, *University of Technology Twintech*, Yemen
- Managed the IT and Multimedia faculties, reporting to the university president.
- Led curriculum development activities for the Business IT and Multimedia faculties.
- Taught courses: *Object Oriented Programming*, *Analysis and Design of Software*, *Human-Computer Interaction*, *Research & Development*.

2011 – 2014 **R&D Engineer / Team Leader / Architect**, *TayalT*, Cairo, Egypt
- Managed multiple products (social media apps and games) and reported to the CEO.
- Coordinated cross-functional teams of technical, marketing, art design, and strategy.
- Designed/maintained the software architectures for the products.
- Developed AI players for games, resulting in improved user ratings and app rankings.

Projects

2021 – 2023 **ROELD Music Toolbox**

Open-source solution for music historians and collectors. *See github.com/n42r/roeld*
- Developed the project based on personal needs of music management.
- Features: Guess release date, generate YouTube playlists from AI-generated tracklist.
- **Results:** Successfully organized and tagged over 50,000 songs.

2016 – 2019 **SIMUTOOL KM Suite**, *SIMUTOOL (EU Project)*

Data cataloging & exchange solution for Industry 4.0. *See github.com/simutool*
- Developed the interfaces, data formats, and the data architecture.
- Managed the whole product and led developers to developed different components.
- **Results:** The project was positively evaluated by an EU panel of experts.

2011 – 2014 **Greetings Studio**, *TayalT*

An App for creating and sharing e-cards and visual sentiments. *See n42r.github.io/gs*
- Technical product management, product design, and single source of truth.
- Led the technical team; maintained the architecture; interfaced with business people.
- **Results:** Number of active users & rating steadily grew to 4 stars on the Apple Store.

2011 – 2013 **Tawla iOS Game**, *TayalT*

Backgammon board game App on the iOS mobile platform. *See n42r.github.io/tw*

- Developed AI players and other components (complex random number generators).
- Led the technical team.
- **Results:** The rating of the App improved from 3 to 4+ stars citing the AI players.

2010 – 2011 **ArtSapiens**, *The German University in Cairo*

Software to compete in the RoboCup 2D Soccer Simulation competition.

- Co-founded the team of 10, selected participants and co-planned the project.
- Designed architecture and led the R&D of complex components (ex., Markov Models).
- **Results:** The team qualified and competed in the 2011 RoboCup finals.

Skills

General: Technical Product Management, R&D, Engineering Design, Data Architecture

Programming: Python, Semantic/Graph Databases, lisp (**Formerly:** Java v5, C++, Prolog, PIC microcontroller programming)

Education

2023 **PhD. (Dr. rer. nat.)**, *Information Systems & Applied Computer Science, Otto-Friedrich-Universität Bamberg*, Bamberg, Germany

Thesis: A data architecture pattern for interoperable cataloging, governance and exchange of datasets (related: data mesh, data space, data lake). *See n42r.github.io/phd*

2010 **MSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

Thesis: A framework for robot perception, including computational and algorithmic models. Ex., if a robot hears a baby but does not see it, is there a baby?

2008 **BSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

Publications

2021 Data-Sharing Markets for Integrating IoT Data Processing Functionalities. *CCF Transactions on Pervasive Computing and Interaction*.

2018 Semantic Data Management for Experimental Manufacturing Technologies. *Datenbank-Spektrum*.

2010 Focused Belief Revision as a Model of Fallible Relevance-Sensitive Perception. *Künstliche Intelligenz 2010: Advances in Artificial Intelligence*.

2010 High-Level Perception as Focused Belief Revision. *European Conference on Artificial Intelligence (ECAI)*.

Languages

Arabic: Mother Tongue

English: Fluent

German: Good understanding, Basic Speaking (B1, Telc Certificate)